

Public Economics and Development

Week 7: Challenges to taxation in weak states: Informal economies and political economy

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

April 14th, 2026

Development of fiscal capacity is a key feature of development

Expansion of government size and fiscal capacity investment in the last century

- Tax share of GDP: 10% in 1910 → 30-50% today
- Tax capacity improved following implementation of broad-based taxes (income tax)
- However, this explains the trend only in developed countries
 - ▶ So economic growth → better fiscal capacity? Not regularly
 - ▶ Separate investment in capacities are key

But in poor states, no noticeable increase in fiscal capacities

- Raise significantly less revenues than developed countries
- Lag behind in the implementation of modern tax technologies
 - ▶ Property registers are manually updated, if updated at all
 - ▶ Do not utilize third-party information that ensures accurate tax information
 - ▶ Less digitalized systems, more loopholes in general

Less capable states rely more on distortionary and narrow based taxes

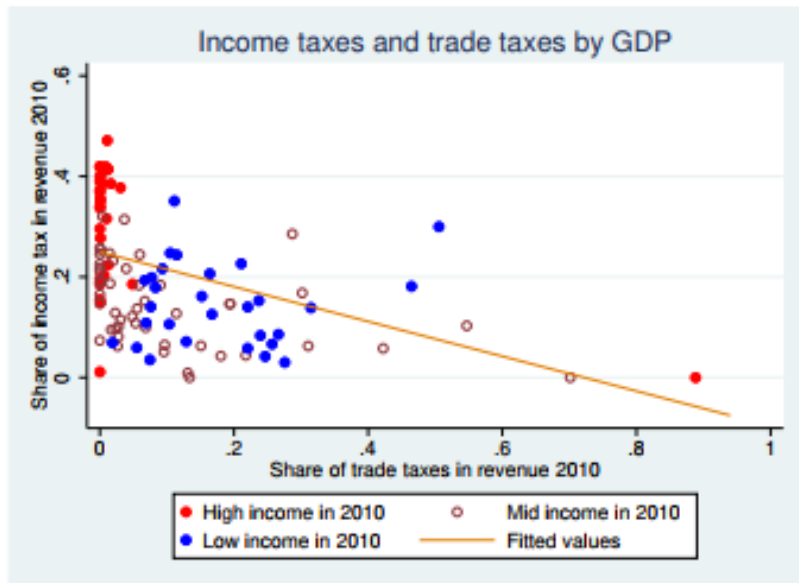
Why do they depend on such taxation

- Broad-base taxes require registries that cover formal sector, and audits
- Narrow-base taxes rely on easily observable bases (trade ports, mines etc)
- Large informal economy and underdeveloped finance sector: Opens up loopholes
- Extractive for select groups by shifting tax burden onto them
- Bad equilibrium: Fiscal capacity $\downarrow \rightarrow$ Public goods and economic development $\downarrow$

What the data tells us

- High income countries rely on broad-based taxes
- Narrow base taxes (tariffs) more observable in low-capacity countries

High income $\rightarrow$ Income tax, Lower income $\rightarrow$ Trade tax



Less capable states suffer from negative shocks from conflicts

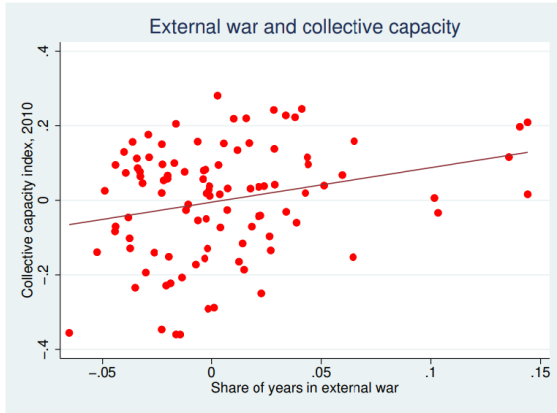
External Conflicts and taxation (Besley, Persson 2010)

- External wars raise value of public goods (e.g. national defense)
- Hence, motivates increases in fiscal capacity

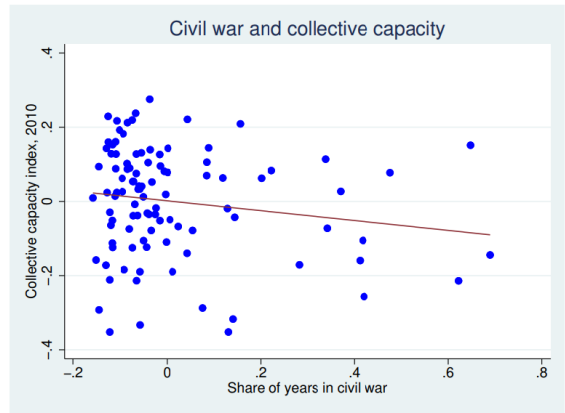
However, many of the global conflicts are internal

- Approx 20-30% of countries worldwide have active civil conflicts during 1960-2000s
- Supermajority of conflicts are intrastate (Ray, Esteban 2017)
- Depletes administrative infrastructure used to mobilize resources
- Hence, leads to decline in tax collection and public goods provision

External wars increase capacity; internal wars hamper it



(a) External wars positively correlate



(b) Internal wars negatively correlate

Roadmap for today

1. Overview setup of the topic

2. Theories on tax leakages

2.1 Allingham, Sandmo (1972): Income Tax Evasion: A Theoretical Analysis

3. Informal economies and taxation

3.1 Gordon, Li (2009): Tax structures in developing countries: Many puzzles and a possible explanation

3.2 Benhassine, McKenzie, Pouliquen, Santini (2018): Does Inducing Informal Firms to Formalize Make Sense? Evidence from Benin

4. Conflicts and implementation of tax authority

4.1 Sanchez de la Sierra (2020): On the Origins of the State: Stationary Bandits and Taxation in Eastern Congo

4.2 Henn, Mugaruka, Ortiz, Sánchez de la Sierra, Wu (2025): Monopoly of Taxation without a Monopoly of Violence: The Weak State's Trade-off from Taxation

Allingham, Sandmo (1972): Income Tax Evasion: A Theoretical Analysis

Q: How and why do some people avoid taxes?

Optimal taxation from the standpoint of the individuals

- Up until now, we assumed that individuals abide by taxes levied on them
- In reality, that is not always the case
 - ▶ Tax avoidance: Exploit legal loopholes to minimize tax liability
 - ▶ Tax evasion: Misreport tax liability illegally
 - ▶ The distinction between the two is *subtle*...

This paper provides a classical model of tax evasion

- Individuals factor in the detection probability to decide on how much income to *report*
- The true income of each individual is given
- If not caught, they keep more income than they would have if they complied to taxes
- If caught, they pay taxes and penalty on difference between true and reported income

Setting up the framework: Individual decisions

Assumptions

- Individual's true income is W
- Tax rate t levied on *reported* income $\bar{W} = W - E$, where E is amount of evasion
- Government conducts tax audit with probability p
 - ▶ We assume audits always catch evasion
 - ▶ If they do not, we can use *effective* probability of detection $p' < p$

Possible scenarios on income

- If not caught: Individuals keep

$$\begin{aligned}Y_N &= W - t\bar{W} = W - t(W - E) \\ &= (1 - t)W + tE\end{aligned}$$

- If caught: Pay taxes on E at rate $\theta > t$

$$\begin{aligned}Y_A &= W - t\bar{W} - \theta(W - \bar{W}) = W - t(W - E) - \theta E \\ &= (1 - t)W - (\theta - t)E\end{aligned}$$

Optimizing on how much to report

Household problem

- We take the expected utility framework

$$V = (1 - p)u(Y_N) + pu(Y_A) = (1 - p)u[(1 - t)W + tE] + pu[(1 - t)W - (\theta - t)E]$$

- Solve the FOC with respect to E

$$[E] : (1 - p)u'(Y_N)t - pu'(Y_A)(\theta - t) = 0$$

$$\iff \frac{u'(Y_N)}{u'(Y_A)} = \frac{p}{1 - p} \frac{\theta - t}{t}$$

- MRS of evaded income = MC of tax evasion (both relative to audited income)

Comparative statics

Comparative statics on p and θ

- Rise in p means audits are more likely to happen and $\frac{\partial E}{\partial p} < 0$
 - ▶ Rise in p raises $\frac{p}{1-p} \Rightarrow \frac{u'(Y_N)}{u'(Y_A)}$ should rise to preserve equality (or Y_N should fall)
 - ▶ Since $Y_N = (1-t)W + tE$ and $Y_A = (1-t)W - (\theta-t)E$ this implies $E \downarrow$
- Rise in θ increases penalty once detected and $\frac{\partial E}{\partial \theta} < 0$
 - ▶ Rise in θ raises $\frac{\theta-t}{t} \Rightarrow \frac{u'(Y_N)}{u'(Y_A)}$ should rise to preserve equality (or Y_N should fall)
 - ▶ With similar logic as above, $E \downarrow$

Comparative statics

Comparative static on tax rate t is a bit more complicated

- Substitution effect: Higher tax increases gains from hiding income (without affecting penalty amount) and raises E
- Income effect: Higher tax makes taxpayer poorer anyway, and if poor are risk averse (of being caught), then they decrease E
- This is because penalty θE does not scale with t
 - ▶ Rising t increases gains from evasion w/o proportionally increasing the cost of detection

Adjust the model: Pay taxes on evasion itself θtE

- Then $Y_A = (1 - t)W + tE(1 - \theta)$ and FOC on tax evasion becomes

$$\frac{u'(Y_N)}{u'(Y_A)} = \frac{p(\theta - 1)}{1 - p}$$

- Here, rise in t raises both gains from evasion *and* penalty if caught
- Thus, there is no substitution effect, and E unambiguously decreases

Higher taxes → Less evasion?

Intuitively, higher taxes increase the burden on taxpayers

- More to gain from hiding income
- Natural expectation: higher $t \Rightarrow$ more evasion

Previous results shows the opposite: higher $t \Rightarrow$ less evasion

- When penalty is on evaded taxes θtE , gains and penalties scale equally with t
- t cancels out of the evasion decision entirely
- Higher t only makes taxpayer poorer, reducing appetite for risky evasion

Empirically, most evidence goes against this theory

- Higher tax rates tend to be associated with more evasion in practice
- Suggests substitution effect dominates in real settings
- Or that penalty structures do not follow the this assumption in practice

What this means in the context of developing countries

Developing countries generally have low detection capacities

- In terms of our model, p is low
- Implying higher E and less tax revenues
- Example: Informal economies (undetected from formal state)

They also may have lower θ

- Lack of legal enforcement on deviant behaviors
- Thus, more evasion (and less tax revenues)
- Example: Less credible threat of enforcing penalties (failure to monopolize tax)

Gordon, Li (2009):
Tax structures in developing countries: Many
puzzles and a possible explanation

Q: Do presence of informal sectors affect choice of tax policies?

Major difference in tax structures across developing and developed countries

- Some could be attributed to social preferences over public goods and redistribution
- But many differences with predicted optimal taxation in generic settings
 - ▶ These taxation preserves production efficiency under plausible assumptions
 - ▶ Rule out tariffs, capital income taxes, differential consumption tax, inflation tax
- Drastic differences in developing countries
 - ▶ Less personal income tax, more tariffs, corporate income tax, and seignorage

This paper proposes that existence of informal sector explains different tax structures

- Incorporates enforcement problems in developing countries due to informal economy
 - ▶ Informal economies: Use cash over financial intermediaries
 - ▶ Taxes on formal economy shifts activities to informal economy → Optimal tax rate ↓
 - ▶ Justifies inflation tax, differential tax across industries, and tariffs

Summary of the difference in tax choices across country income

Table 1

Sources of government revenue (1996–2001).

GDP per capita	Tax revenue (% of GDP)	Income taxes (% of revenue)	Corporate income tax (% of income taxes)	Consumption and production taxes (% of revenue)	Border taxes (% of revenue)	Inflation rate	Seignorage income (% of revenue)	Informal economy (% of GDP)
<\$745	14.1	35.9	53.7	43.5	16.4	10.6	21.8	26.4
\$746–2975	16.7	31.5	49.1	51.8	9.3	15.7	24.9	29.5
\$2976–9205	20.2	29.4	30.3	53.1	5.4	7.4	6.0	32.5
All developing	17.6	31.2	42.3	51.2	8.6	11.8	16.3	30.1
>\$9,206	25.0	54.3	17.8	32.9	0.7	2.2	1.7	14.0

Notes: Authors' calculations based on available data between 1996 and 2001 from Government Finance Statistics (IMF, 2004a), International Finance Statistics (IMF, 2004b), and World Development Indicators (World Bank, 2003). The ranges for GDP per capita follow the World Bank 2003 classification of low income, lower middle income, middle income and high income countries. Seignorage is measured as the increase in reserve money and currency in circulation. Estimates of the size of the informal economy in 1999 in Column (9) are from Friedrich Schneider (2002), who uses the currency demand approach in estimation. Data within each cell are weighed averages. Tax revenue (% of GDP), inflation rate, and the size of the informal economy (% of GDP) are weighed by GDP of each country. Corporate income tax (% of income taxes) is weighed by the total income tax revenue of each country. All other data are weighed by the tax revenue of each country.

- High corporate income tax, tariffs, seignorage in developing countries
- Less personal income tax
- Statutory tax rates are similar across different countries!

Tax policy with limited information: Differential excise taxes

Characterizing informal economy into a model

- Informal economy not observed by government
 - ▶ Thus, taxes cannot be levied to such firms
 - ▶ So profits for such firms are revenues - wage cost - capital costs
- Formal firms use financial systems and have higher outputs, but subject to tax
- Firms become formal if net profits exceed those from informal economy
 - ▶ High excise taxes reduce the net profits → tax revenue drops

Variation across industries

- Sectors with low increase in output from formalizing → lower tax rates than desired
- This results in distortion in domestic production, which are offset by tariffs on imports
- Inflation tax and red tape on lightly taxed sectors may be attractive

How to increase benefits of using financial sector

Determinants of benefits to using financial sectors

- Capital intensive firms: They need lending to finance purchase of capital
- If firms operate at wide distance, financial sector reduces the cost
- High revenue per employee: More cash transaction exposes to internal theft
 - ▶ Service sectors tend to be less capital intensive and easily transfer to informal economy
- Firms that do not formalize avoid corporate and excise tax, but pay inflation tax
- Thus, taxes are kept low for non capital-intensive sectors
 - ▶ Capital-intensive ones more likely to stick to formal even without intervention
 - ▶ This is meant to shift informal economies to formal ones
- This explains high capital income tax in developing countries (vs. income tax)
- Inflation taxes also raise cost of keeping cash, inducing firms to formalize

Motivating uses of other anti-competitive measures

Tariffs

- Higher tariff protection for capital intensive sectors
 - ▶ Meant to offset distortions due to uneven domestic tax enforcement
 - ▶ Specifically, aim is to induce production to a sector more likely to formalize

Other anti-competitive measures

- Provide monopoly power to capital-intensive firms
- Impose tighter regulation or monitoring on informal firms
 - ▶ Even if revenue from such firms cannot be observed, other activities could be
 - ▶ Higher red tape measure on those firms as monitoring device

Key takeaways

Existence of informal firms twist the choice of optimal taxation

- More capital taxation, tariffs, and inflation taxes
- This is determined by the returns to using financial sectors
 - ▶ If financial sectors provide benefits, less incentives to join informal economy
 - ▶ As developing countries have less developed finance sectors, high informality
- Thus, models should incorporate possibility of informal economies

However, leaving informal economies as is is undesirable

- Weakens capacity to provide essential public goods
- Resource misallocation $\uparrow$ and innovation $\downarrow$ (cost of hiding leave little for innovation)
- Thus, efforts to induce firms to formalize through third-party information, legal measures, and expansion of electronic filing (Benhassine et al 2018)

Benhassine, Mckenzie, Pouliquen, Santini
(2018):
Does Inducing Informal Firms to Formalize
Make Sense? Evidence from Benin

Q: Can governments induce informal firms to formalize?

Many small firms in the developing country are informal

- In some cases, their estimated size is equivalent to 70% of GDP
- These could be very problematic
 - ▶ Informal firms cannot access formal finance sector, public contracts etc
 - ▶ Lost tax revenues for the government
- Thus, reforms to mandate business entry regulation is in place in many places

This paper uses randomized experiment to see what can be done to induce formalization

- Involves 3,600 firms in Benin
- Introduction of entrepreneur legal status, a simplified formalization regime
- Estimates benefits for firms and costs of the intervention
- Also provides suggestion on who is most likely to formalize

Institutional Context in Benin

Entrepreneur status reform in Benin

- Came into effect in 2011
- Aim to facilitate formalization through simplified procedures
- Costs also reduced by 70-80%
- As these were rolled out suddenly, many firms seemed to have been unaware

What happens when firms formalize (vs informal)

- Get tax ID and exempt for taxation for one tax year (and 3-year reduced taxation)
- Informal firm addresses unknown to govt (so difficult to collect tax)

Experiment design

Treatment across 3595 firms

- A: In-person visits with verbal explanation and assistance in paperworks
- B: Help open a bank account and provide training in return for formalizing
- C: Provide tax mediation service and tax preparation service
- Treatment1 got A, Treatment2 got A+B, Treatment 3 got A+B+C
- The experiment data is merged with admin data on formalization and firm surveys

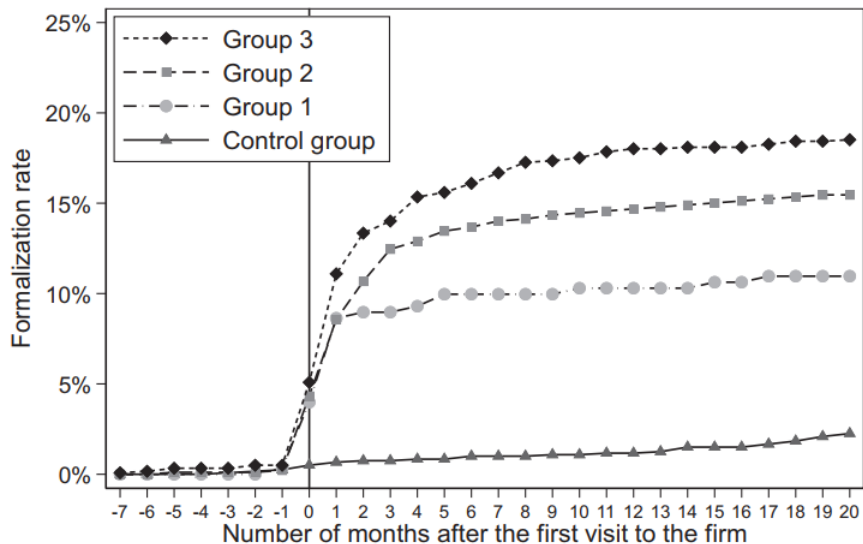
The regression follows an OLS format, given randomized assignment

$$y_i = \alpha + \beta_1 T1_i + \beta_2 T2_i + \beta_3 T3_i + X_i + e_i$$

- Follow-up surveys: Form a panel of firm-years

$$y_{it} = \alpha + b_1(T1_i \cdot F1) + b_2(T2_i \cdot F1) + b_3(T3_i \cdot F1) \\ + c_1(T1_i \cdot F2) + c_2(T2_i \cdot F2) + c_3(T3_i \cdot F2) + \pi Y_{i0} + \gamma M_{i0} + X_i \delta + e_i$$

More firms to formalize



Notes: N=3,596

But limited effect on take-up of further programs

	(1)	(2)	(3)	(4)	(5)
	Has a bank account ^B	Loan contracted in 2014-16 ^B (bank or MFI)	Attended business training in the past year	The firm does any form of accounting ^B	Share of business practices implemented (26 questions)
1st stage: impact of treatment allocation:					
Group1 X year1 (b1)	0.028 (0.031)	- 0.030 (0.024)	0.008 (0.018)	- 0.099*** (0.026)	- 0.046*** (0.012)
Group2 X year1 (b2)	- 0.008 (0.018)	- 0.018 (0.014)	0.081*** (0.014)	0.007 (0.017)	- 0.004 (0.008)
Group3 X year1 (b3)	0.017 (0.016)	- 0.009 (0.013)	0.112*** (0.012)	0.023 (0.015)	0.004 (0.008)
Group1 X year2 (c1)	0.054* (0.031)	0.006 (0.025)	0.023 (0.019)	- 0.052* (0.028)	- 0.018 (0.015)
Group2 X year2 (c2)	0.011 (0.019)	0.051*** (0.016)	0.113*** (0.015)	0.020 (0.018)	- 0.005 (0.009)
Group3 X year2 (c3)	0.003 (0.016)	0.015 (0.014)	0.145*** (0.013)	0.047*** (0.016)	0.007 (0.008)

And limited impact on firm performance

Table 5

Impact on Firm Performances.

	(1)	(2)	(3)	(4)	(5)
	Total sales in the last day ^{αβ} (CFAF)	Total sales in the last week ^{αβ} (CFAF)	Last month profit ^{αβ} (CFAF)	Summary index of sales and profit ^{αβ}	Total number of employees ^α
1st stage: impact of treatment allocation:					
Group1 X year1 (b1)	2228 (2754)	12,496 (14,029)	- 8053* (4798)	0.008 (0.057)	- 0.22** (0.10)
Group2 X year1 (b2)	540 (1451)	- 7376 (7312)	- 3016 (3021)	- 0.052* (0.031)	- 0.06 (0.09)
Group3 X year1 (b3)	- 114 (1384)	- 1224 (6399)	- 3106 (2858)	- 0.010 (0.030)	- 0.11 (0.08)
Group1 X year2 (c1)	602 (2930)	12,192 (14,243)	470 (5742)	0.041 (0.060)	- 0.09 (0.10)
Group2 X year2 (c2)	1246 (1832)	- 5235 (8010)	- 874 (3377)	- 0.007 (0.036)	0.05 (0.07)
Group3 X year2 (c3)	1847 (1669)	3998 (7911)	242 (3233)	0.026 (0.035)	0.08 (0.07)

Cost-benefit analysis and takeaways

Cost-benefit analysis

- Total costs in the 2 years: Equivalent to US\$ 620,000
- Costs per formalization: Range from \$1,237 – \$ 2,217
- This is more than 10 times of the profits from formalization
- Recouping costs through taxes: Takes estimated 20 years
- Better targeting could help: More educated firmowners more likely to formalize

Takeaways

- Formalization can bring in additional tax revenues in theory
- But implementation is costly: Takes time to recoupe costs of additional interventions
- What now? Target those who are more likely to formalize?

Sanchez de la Sierra (2020):
On the Origins of the State: Stationary Bandits
and Taxation in Eastern Congo

Q: How do stationary bandits mimic essential functions of the state?

Monopoly of violence, taxation, and law enforcement is an essential function of state

- But even in developed countries, these are modern concepts
- In what environment do these functions emerge?
- However, many quantitative data exist *since* the states are formed
- so what explains *the birth* of states?

This paper investigates this question using a natural experiment

- Uses Democratic Republic of Congo (DRC) as a context
 - ▶ Vacuum of the formal state presence (DRC lost de facto control of part of its territory)
 - ▶ Leaving armed actors to monopolize violence, taxation, and other essential services
- Motives of armed actors determined by external changes in resource prices
- Rise of state function related to which types of resource shock is in place

Conceptualizing state functions

When are states likely to emerge?

- States monopolize violence to repress threats to the property of the governed
- They then monopolize taxation to finance their operations
- The potential tax revenues motivate armed actors to protect welfare of the population

Motivating hypotheses

- When do state emerge and develop state capacity?
 - ▶ Likely when an output that can be easily taxed exists
 - ▶ The output themselves should promise significant returns for the armed actors
- What is the net benefit for the population?
 - ▶ If investments spurred by protection $>$ disincentives created by taxing production
 - ▶ If taxing output (suppliers) is difficult, states may choose to target consumption
- In production economies: State emerges with high output prices at production sites
- In consumption economies: State emerges where the outputs are being consumed

Setting and data in DRC

Setting: Eastern DRC in early 2000s

- Mineral extraction in eastern DRC (lack of formal state) promises high returns
- Many militias (regional, external actors, DRC army) have incentives to take over
- Demand shock for different minerals
 - ▶ Coltan: Heavy, demand shock driven by announcement of PS2
 - ▶ Gold: Light/concealable, safe haven of values during recession

Data and analysis

- Mining site and local history (outputs) compiled by surveyors on the ground
- Uses the following municipality-year level regression

$$Y_{jt} = \beta_t + \alpha_j + \gamma_{\text{coltan}} C_j p_{\text{coltan},t}^{\text{US}} + \gamma_{\text{gold}} G_j p_{\text{gold},t}^{\text{US}} + e_{jt}$$

State emerges differently in production and consumption economies

VARIABLES	MUNICIPALITY ATTACKED (1)	MINE					VILLAGE		
		Customs Tax (2)	Entry Fees (3)	Stationary Bandit (4)	Security Service (5)	Extensive-Margin Index (6)	Stationary Bandit (7)	Security Service (8)	Extensive-Margin Index (9)
Coltan _j × p_{ct}	.15*** (.03)	.06** (.02)	.01 (.02)	.07*** (.03)	.06** (.02)	.16*** (.06)	.04 (.03)	.02 (.02)	.07 (.05)
Gold _j × p_{gt}	-.02 (.03)	-.04 (.02)	.04* (.03)	.05 (.04)	.05 (.04)	.10 (.07)	.12*** (.04)	.15*** (.04)	.33*** (.08)
Observations	4,158	4,046	4,052	3,991	4,032	3,903	4,302	4,302	4,302
R ²	.39	.59	.69	.60	.62	.65	.50	.42	.50

Coltan: Benefits locations without stationary bandits

Variables	Savings Index (1)	No. of Weddings (2)	No. of Immigrants (3)	Welfare Index (4)
A. Coltan Price (C_j)				
$C_j \times p_{c,t}$	.08* (.05)	.23*** (.08)	.54*** (.16)	.36*** (.08)
$C_j \times p_{c,t} \times SB_{jt}$	-.01 (.01)	.01 (.02)	.01 (.04)	.02 (.02)
$C_j \times p_{c,t} \times SB_{jt} \times Army_{jt}$	-.01 (.01)	.02 (.02)	.03 (.05)	.03 (.03)
$C_j \times p_{c,t} \times SB_{jt} \times Militia_{jt}$	.02** (.01)	.08*** (.02)	.17*** (.04)	.11*** (.03)
Observations	3,582	3,466	3,523	2,669
R^2	.36	.54	.40	.52

Gold: Benefits locations with stationary bandits

	B. Gold Price (G_j)			
$G_j \times p_{g,t}$	-.08 (.05)	-.01 (.09)	-.07 (.14)	-.06 (.09)
$G_j \times p_{g,t} \times SB_{jt}$	-.00 (.01)	-.02 (.01)	-.04* (.02)	-.01 (.01)
$G_j \times p_{g,t} \times SB_{jt} \times Army_{jt}$	.00 (.01)	.04** (.02)	.09*** (.02)	.04*** (.01)
$G_j \times p_{g,t} \times SB_{jt} \times Militia_{jt}$	.00 (.01)	.03** (.01)	.08*** (.02)	.03** (.01)
Observations	3,582	3,466	3,523	2,669
R^2	.36	.53	.39	.50

Takeaways and discussion

Summaries of key findings

- Armed groups sometimes form monopoly of violence, taxation, and protection
- Especially if outputs with high returns that can be expropriated are present
- In turn, they form essential state functions that sometimes benefitting the well-being

Remaining questions

- Sometimes the population would rather support armed groups over the government
 - ▶ e.g. Mexico, Colombia, or any places where organized criminal groups provide services
- But that is not always the case
- So how do armed groups legitimize their presence?

Henn, Mugaruka, Ortiz, Sánchez de la Sierra,
Wu (2025):

Monopoly of Taxation without a Monopoly of
Violence: The Weak State's Trade-off from
Taxation

Q: Tradeoffs between monopoly of taxation & monopoly of violence?

Weak states often fail to exert de facto control over their jurisdictions

- Non-state actors (rebels) take over, who sometimes provide stability over
- If these areas are economically valuable, state loses lot of fiscal revenue
- So is it beneficial for state to take formal authority over those lands?
 - ▶ Regain lost fiscal revenues
 - ▶ But could trigger inadvertent response from rebels (e.g. plundering)

This paper investigates the effect of state-building attempts to increase tax revenues

- Military powers to reclaim land and replace rebels
- Negotiate with rebels (often with concession to the rebels)
- Leverage subnational evidence in Democratic Republic of Congo (DRC)

State-building efforts to drive rebels out of DRC

State absence in Eastern DRC (and postcolonial Africa in general)

- Many parts of DRC (East) were under armed group control since 1998
- Most notable of these organizations is FDLR (Forces de Libération du Rwanda)

Military operation and peace treaty with rebels

- Largest peace treaty: Sun City peace treaty in 2004
 - ▶ Various armed groups integrated into Congolese army
 - ▶ Allowed government to regain half the country without triggering violence
 - ▶ Other peace agreements had smaller goals and lower credibility
- Largest military operation: Kimia II (March 2009)
 - ▶ Joint operation with UN, national army, and Rwanda
 - ▶ Targeted FDLR presiding and taxing over Basile Chiefdom region since 2005
 - ▶ Although defeated, FDLR launched theft operations from the nearby hilly forest region

Data and empirical strategy

Constructing the data

- Armed group presence: village-level data through interviews with former combatants
 - ▶ Perpetrators' affiliation, motivation (pilgrage/retaliate/etc), actions taken

Effects of Kimia II and Sun City Peace Treaty

- Kimia II: FDLR vs non-FDLR villages pre ('05-'09) and post ('10-'12) in Basile Chiefdom
- Peace treaty: Integrated villages vs others in the region before and after '02

$$Y_{i,t} = \alpha_i + \beta_t + \sum_{k=-4}^{k=3} \beta_k \text{Treat}_i \times 1[t = t_0 + k] + e_{i,t}$$

- ▶ $t_0 = 2009$ for Kimia II effects, 2002 for peace treaty effects

Kimia II: FDLR increased violence afterwards

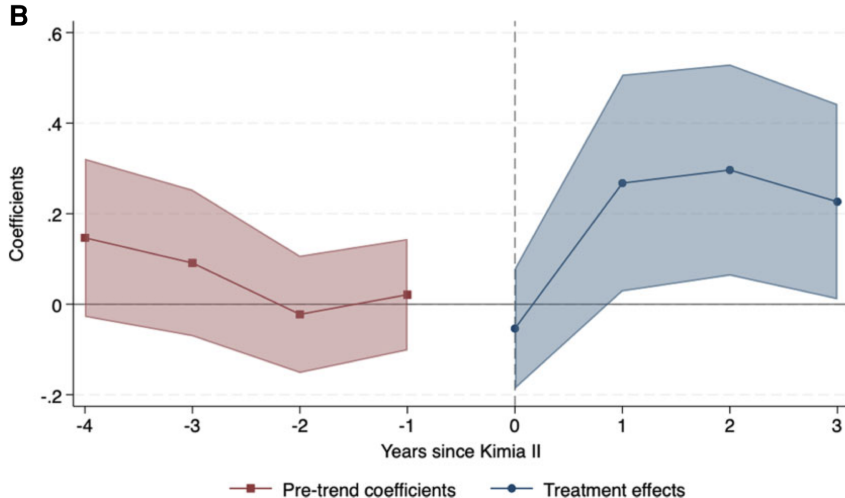


FIGURE 4

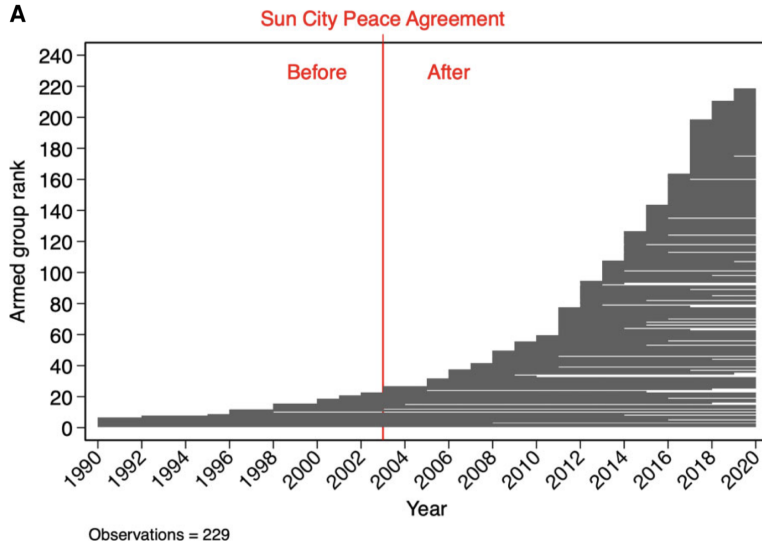
Kimia II campaign and FDLR violence, for FDLR state villages and the rest (A) Times Series and (B) Event Study

Kimia II: Driven by increase in pillages by FDLR

TABLE 1
Mechanism—purported motivation for the attacks

	<i>Dependent variable: FDLR attack</i>						
	Any (1)	Conquest (2)	Punishment (3)	Pillage (4)	Conquest (5)	Punishment (6)	Pillage (7)
$FDLR_i \times 1(t > 2009)$	0.240*** (0.068)	−0.001 (0.018)	0.043*** (0.006)	0.220*** (0.063)	−0.030 (0.020)	0.037*** (0.006)	0.210*** (0.060)
Control FDLR Conquest	N	N	N	N	N	Y	Y
Control FDLR Punishment	N	N	N	N	Y	N	Y
Control FDLR Pillage	N	N	N	N	Y	Y	N
Observations	3,474	3,474	3,474	3,474	3,474	3,474	3,474
R^2	0.14	0.08	0.08	0.13	0.29	0.28	0.17
Mean Dep. Var.	0.07	0.02	0.02	0.04	0.02	0.02	0.04
Village clusters	193	193	193	193	193	193	193
Chiefdom-year Clusters	360	360	360	360	360	360	360
Conley (1999) p -value	0.00	0.97	0.00	0.00	0.12	0.00	0.00

Peace treaty: Rise of new armed groups since treaty



Takeaways and discussions

What (doesn't) explain the effects

- Actual violence increased (as opposed to *reported* violence)
- Violence by other actors do not increase following Kimia II
 - ▶ Thus when incentive to protect villages (for tax revenues) vanish, violence increase
- Treaty raised revenues, but limited commitment ability led to questioning of legitimacy

Takeaways and questions

- Kimia II: Monopolize violence (rebels out), but lost taxation due to ensuing violence
- Peace treaty: Expand tax revenues but no longer monopolize violence
- If faced with these trade-offs what should the states do?
 - ▶ Asserting force desirable if state has more capacity to prevent crime
 - ▶ Bargaining potentially desirable if states can commit to not negotiate with future rebels and have resource to integrate rebels

Unanswered questions

Informal economies reduce potential for tax revenues

- Thus, states have less information about tax payers
- This induces developing countries to choose different types of taxes
 - ▶ Those that are easily detectable and difficult to avoid
 - ▶ Trade taxes, corporate income tax, consumption tax
- But formalizing them is also costly
- Then, how to increase amount of information (e.g. new technologies?)

Failure to monopolize taxes

- States sometimes compete with non-state actors (armed groups / local chiefs)
- Tax administration is limited in scope
- Any way to incorporate local knowledge?